EAS 487
HOMEWORK 4
SOLUTIONS

1. $\langle E \rangle = \frac{1}{\pi (D/2)^2} \int_0^{2\pi} \int_0^{D/2} \cos^n (\pi r / D)\, r\, dr\, d\theta$ ← do this generally

$= \frac{8}{D^2} \int_0^{D/2} \cos^n (\pi r / D)\, r\, dr \qquad x = \frac{\pi}{D} r \quad dx = \frac{\pi}{D} dr$

$= \frac{8}{\pi^2} \int_0^{\pi/2} \cos^n (x)\, x\, dx \quad = 1,\ 0.463,\ 0.297$ for $n = 0, 1, 2$

$\langle E^2 \rangle = \frac{8}{\pi^2} \int_0^{\pi/2} \cos^{2n} (x)\, x\, dx \quad = 1,\ 0.297,\ 0.172$ likewise

a) $\epsilon_{ap} = \frac{|\langle E \rangle|^2}{\langle |E|^2 \rangle} = 1,\ 0.72,\ \underline{0.51}$

b) $D = \frac{4\pi}{\lambda^2} A_{phys}\, \epsilon_{ap} = 1.75 \times 10^5\, \epsilon_{ap} = 52.4\ dB,\ 51\ dB,\ \underline{49.5\ dB}$

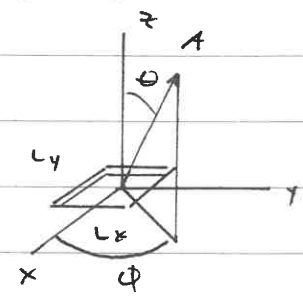
↳ $\Omega = \frac{4\pi}{D} \approx (HPBW)^2 \rightarrow HPBW = 0.48°,\ 0.57°,\ \underline{0.68°}$

2. $$F(k_x, k_y) = \int_{-L_x/2}^{L_x/2} \int_{-L_y/2}^{L_y/2} e^{j(k_x x' + k_y y')}\, dx'\, dy'$$

$$= L_x L_y \frac{\sin k_x L_x/2}{k_x L_x/2} \frac{\sin k_y L_y/2}{k_y L_y/2} = L_x L_y \frac{\sin X}{X} \frac{\sin Y}{Y}$$

$$X = \frac{k_x L_x}{2} = \frac{2\pi}{2\lambda} 50\lambda \sin\theta \cos\phi = 50\pi \sin\theta \cos\phi$$

$$Y = \frac{k_y L_y}{2} = \frac{2\pi}{\lambda} 50\lambda \sin\theta \sin\phi = 50\pi \sin\theta \sin\phi$$



$$|F|^2 = \left(\frac{\sin X}{X}\right)^2 \left(\frac{\sin Y}{Y}\right)^2$$ NORMALIZED

RADIATION ONLY IN $z > 0$ SPACE

SO $P_{max} = |F|^2 = 1$ $\qquad P_{avg} = \frac{1}{4\pi} \underbrace{\int_0^{2\pi} \int_0^{\pi/2} |F|^2 \sin\theta\, d\theta\, d\phi}_{P_{total}}$: 4×10^{-4} NUMERICALLY

$$D = \frac{P_{max}}{P_{avg}} = 4\pi \frac{P_{max}}{P_{total}} = \frac{4\pi}{\lambda^2} A_{eff}$$

SO $\frac{P_{max}}{P_{total}} = 2500 = \frac{A_{eff}}{\lambda^2}$ $\qquad A_{eff} = (50\lambda)^2 = A_{phys}$

3. THE APERTURE DISTRIBUTION IS UNIFORM IN THE Y DIRECTION FOR $|y| < \frac{L}{2}$, AND THE ANGULAR SPECTRUM IN THE Y-Z PLANE IS THEREFORE

$$|F|^2 = \left|\frac{\sin Y}{Y}\right|^2 \qquad Y = \frac{\pi}{\lambda} L \underbrace{\sin\theta}\, \overbrace{\sin\phi}^{1 \text{ IN Y-Z PLANE}}$$

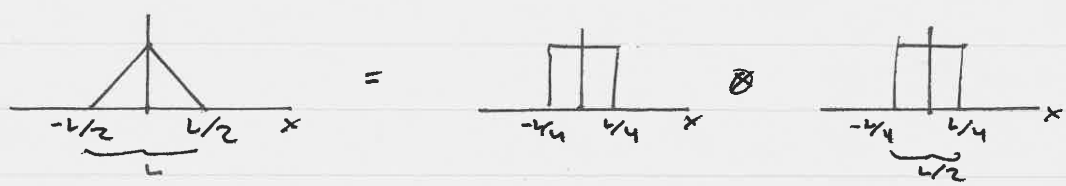
$\theta_{1/2}$ OCCURS WHEN $\left|\frac{\sin Y}{Y}\right|^2 = \frac{1}{2} \Rightarrow Y \cong 1.4 = \frac{\pi L}{\lambda}\sin\theta$ IN Y-Z PLANE

OR $\sin\theta_{1/2} = 1.4\,\frac{\lambda}{\pi L}$ ASSUME $L \gg \lambda$

$$\theta_{1/2} = 0.445\,\frac{\lambda}{L}$$

$$\text{HPBW} = 2\theta_{1/2} = 0.89\,\frac{\lambda}{L} \Rightarrow \underline{\sim 1^\circ \text{ FOR}} \; L = 50\lambda$$

MEANWHILE, THE APERTURE DISTRIBUTION IN X IS TRIANGULAR:



THE ANGULAR SPECTRUM ASSOCIATED WITH EITHER RECTANGULAR DIST IS

$$|F|^2 = \left|\frac{\sin X}{X}\right|^2 \qquad X = \underbrace{\frac{\pi}{\lambda}\frac{L}{2}}\sin\theta\, \overbrace{\cos\phi}^{1 \text{ IN X-Z PLANE}}$$

SO THE ANGULAR SPECTRUM FOR THE TRIANGLE (BY CONVOLUTION THEOREM) IS

$$|F|^2 = \left|\frac{\sin X}{X}\right|^4 \; ; \quad \text{FIND } \theta_{1/2} \text{ FROM } \left|\frac{\sin X}{X}\right|^4 = \frac{1}{2}$$

$$\Rightarrow X \cong 1 = \frac{\pi}{\lambda}\frac{L}{2}\sin\theta_{1/2}$$

$$\theta_{1/2} \approx \frac{2}{\pi}\frac{\lambda}{L} = 0.64\,\frac{\lambda}{L}$$

$$\text{HPBW} = 2\theta_{1/2} = 1.27\,\frac{\lambda}{L} \Rightarrow \underline{1.46^\circ} \text{ FOR } L = 50\lambda$$

SO THE MAIN BEAM IS BROADENED BY THE TAPER

WHAT ABOUT SIDELOBES ?

IN Y-Z PLANE, GET FIRST SIDELOBE WHEN $Y = \pm \frac{3\pi}{2}$

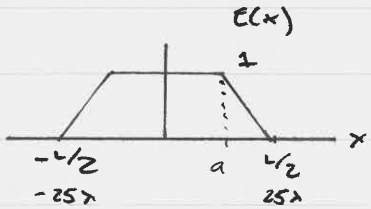
$$|F|^2 = \left|\frac{\sin Y}{Y}\right|^2 = \left(\frac{1}{3\pi/2}\right)^2 = 0.045 = \underline{-13.5\ dB}$$

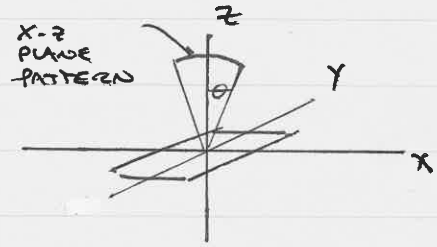
IN X-Z PLANE, ALSO GET 1ST SIDELOBES WHEN $X = \pm \frac{3\pi}{2}$

$$|F|^2 = \left(\frac{\sin X}{X}\right)^4$$

$$= \left(\frac{1}{3\pi/2}\right)^4 = 0.002 = \underline{-27\ dB}$$

b)





$$F(k_x) = \int_{-L/2}^{L/2} \underbrace{E(x)}_{\text{EVEN FCN}}\ \underbrace{e^{jk_x x'}}_{\cos k_x x' + j\underbrace{\sin k_x x'}} dx' \qquad k_x = \frac{2\pi}{\lambda}\sin\theta \text{ IN X-Z PLANE}$$

ODD FUNCTION → WON'T SURVIVE INTEGRAL

$$= 2\int_0^{L/2} E(x)\cos \underbrace{k_x}_{\frac{2\pi}{\lambda}\sin\theta} x'\, dx'$$

CHANGE VARIABLES : $\frac{2\pi}{\lambda}x' = x$

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$$= \underbrace{\frac{\lambda}{\pi}}_{\text{ARBITRARY}} \int_0^{50\pi} E(x)\cos(x\sin\theta)\,dx \qquad x' = \frac{L}{2} \rightarrow x = \frac{2\pi}{\lambda}\frac{L}{2} = 50\pi$$

$$\propto \int_0^a \cos(x\sin\theta)\,dx + \int_a^{50\pi}\left(\frac{50\pi - x}{50\pi - a}\right)\cos(x\sin\theta)\,dx$$

(change back variables)

$|F|^2$ PLOTTED VS. θ → I FIND $a = 53 \cdot \frac{\lambda}{2\pi} \sim 8.43\lambda$

(FOR -20 dB SIDELOBES) OR ABOUT $\underline{\underline{1/6}}$ OF WAY TO EDGE

$\theta_{1/2} = 0.0119$ RAD NUMERICALLY

$HPBW = 2\theta_{1/2} = \underline{\underline{1.37^\circ}}$ (MORE THAN RECTANGLE, LESS THAN TRIANGLE !)

4. FRESNEL ZONE CALCULATION

$L_x = 100\,m$, $\lambda = 1\,m$ $\Rightarrow$ $\frac{2D^2}{\lambda} = 20\,km$ EVALUATE PATTERN

AT $r = 2\,km$, $200\,km$

$$f \propto \int_{APERTURE} J(x') \, e^{j(k_x x' - k x'^2/2r)} \, dx'$$

WHERE $k_x = k \sin\theta$

$$f(\theta) \propto \underbrace{\int_{-L/2}^{L/2} e^{jk(x' \sin\theta - x'^2/2r)} \, dx'}$$

WRITE $k = \frac{2\pi}{1m}$

HAS REAL & IMAG PARTS

ie $\cos(\quad) + j\sin(\quad)$

$F^2(\theta) \propto (\text{REAL PART})^2 + (\text{IMAGINARY PART})^2$ SEE PLOTS